

PETERS ELIJAH TEMIDAYO

Electrical & Electronics Engineer | AI/ML & Hardware Co-Simulation

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PROFESSIONAL SUMMARY

Electrical & Electronics Engineer specializing in software–hardware co-simulation, electronic circuit design, and machine learning. Skilled in validating integrated engineering workflows and AI model outputs using Python, LTspice, KiCad, and MATLAB. Analytical and self-directed, with a track record of taking projects from design through to hardware validation, and a strong interest in building intelligent AI and hardware systems.

EDUCATION

B.Eng, Electrical & Electronics Engineering | [Olabisi Onabanjo University](#) 2021 – 2026
Second Class (Upper Division) — Ogun State, Nigeria

Relevant coursework: Circuit Analysis, Software–Hardware Co-simulation, Machine Learning, Power Electronics, Control Systems, Digital Signal Processing, Computational Engineering Methods.

West African Senior School Certificate Examination (WASSCE) | [Suzliz Private School](#) 2017 - 2020
Ogun State, Nigeria

EXPERIENCE

Electronics Circuit Design Expert | [Micro1](#) May 2026 – Present

- Execute software–hardware co-simulation tasks, validating electronic schematics, PCB layouts, and netlists in LTspice, KiCad, and DesignSpark PCB.
- Design, test, and troubleshoot circuit architectures including flyback converters and transformer phase matching, ensuring adherence to real-world physics and EDA standards.
- Evaluate and correct AI-generated engineering outputs, identifying edge cases and technical hallucinations to refine model reasoning in hardware logic.
- Develop evaluation rubrics and execution tasks for circuit-logic workflows, improving the accuracy of specialized engineering AI models.

AI Technical Trainer & Data Specialist | [Micro1](#) Nov 2025 – Feb 2026

- Analyzed and labeled complex datasets to improve LLM accuracy in code generation and logical reasoning.
- Led data-collection initiatives for technical workflows, ensuring high-fidelity inputs for developer-focused AI models.
- Validated technical scripts and documentation, maintaining a high accuracy rate while identifying edge cases in model responses.

Data Science Intern | [Codsoft](#) Aug 2024 – Sept 2024

- Built and trained machine learning models on real-world datasets using Python, Pandas, and Scikit-Learn.
- Performed data cleaning, visualization (Matplotlib/Seaborn), and feature engineering to optimize model performance.

Data Science Intern | [DSN-OOU](#) Mar 2024 – May 2024

- Conducted exploratory data analysis (EDA) on diverse datasets to generate actionable insights.
- Applied statistical methods and data visualization using Pandas, NumPy, and Seaborn.

PROJECTS

AuraPass — Real-Time AI-Powered Access Control System | [Final-Year Project](#) 2026

- Designed and validated a real-time, AI-powered access-control system using software–hardware co-simulation for secure university exam verification.
- Engineered a deep-learning biometric pipeline with Python, OpenCV (Haar Cascades), and DeepFace (FaceNet) to extract 128-byte facial embeddings and perform real-time cosine-distance matching.
- Developed a constraint-satisfaction algorithm to parse bulk student datasets (CSV) and allocate exam seating, preventing adjacent duplicate-course placements.
- Built a secure hardware bridge with bidirectional serial communication between a Tkinter GUI and an ATmega328P microcontroller, driving a power-isolated 12V motor actuation circuit simulated in Proteus.

TECHNICAL SKILLS

Languages & Scripting: Python, MATLAB, SQL, C++, HTML, CSS, Bash

ML & Data: OpenCV, MediaPipe, DeepFace, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, SMOTE

Engineering & Simulation: LTspice, KiCad, DesignSpark PCB, Ansys (SpaceClaim/Fluent), Simulink, Proteus

Core Competencies: Machine Learning, Computer Vision, Software–Hardware Co-simulation, Circuit Design, Predictive Modeling, EDA

Soft Skills: Autonomous Execution, Analytical Problem-Solving, Technical Communication, Time Management, Adaptability

CERTIFICATIONS

- Data Analyst Certification — Micro1, 2026
- Data Science & Machine Learning Virtual Training — Codsoft, 2024
- Artificial Intelligence & Data Science Bootcamp — DSN-OOU, 2024